

Information as the Uncollapsed State: An Observer-Relative Ontology and the Identity of Asymmetry and Entropy

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Abstract

This paper argues that the boundary between the "physical" and the "informational" is not a feature of the world but an artifact of the observer who draws it. It proposes that physics exists in exactly two states relative to any observer — collapsed (already resolved through interaction) and uncollapsed (still potential) — and that what is called information is the uncollapsed state described from within. On this reading, information asymmetry and thermodynamic entropy are not analogous quantities joined by metaphor: they are one observer-relative measurement under two vocabularies, since entropy is itself defined relative to the macrostate an observer can distinguish. The paper extends the identity to reference and meaning, arguing that every act of pointing is executed by a physical nervous system paying a thermodynamic cost, and that deceptive signaling — camouflage, mimicry, false reference — is therefore not an anomaly for an informational ontology but a prediction of it. Finally, it derives a methodological consequence with empirical content: an instrument built to read the geometry of the uncollapsed should read structure and remain structurally silent on outcome. This prediction is examined against the author's own adjudicated results, including those that went against the framework.

Keywords: information asymmetry, observer-relative entropy, thermodynamics of information, philosophy of information, deceptive signaling, Umwelt, statistical manifolds

1. The Line That Does Not Exist in the World

There is a habit so old that it is mistaken for perception: the world is split in two. On one side, the real — matter, energy, the things that resist when touched. On the other, the abstract — ideas, theories, language, information: things that appear to float above the real, describing it without being it.

The argument advanced here is that this line does not exist in the world. It exists in the observer who draws it. Once removed, a set of problems that appear mysterious — about markets, about attention, about why a system that processes information degrades faster than the information it carries — cease to be mysterious and become physical.

This postulate underlies a programme of empirical work in financial series and in the human oculomotor system [1, 2]. The postulate itself has rarely been stated on its own. That is the purpose of this paper.

2. One Substance, Two States

The strong form of the claim is the only one worth examining. There is one thing: physics — the configuration of matter and energy in spacetime. What is called abstraction, and what is called

information, is not a second kind of substance layered upon it. It is the same physics in a different state relative to an observer.

That state takes two values. **Collapsed**: physics a given observer has already resolved — the transaction that executed, the photon that interacted, the event that occurred. **Uncollapsed**: physics that, from that observer's vantage point, has not resolved — the field of what remains potential.

Two established results place this beyond metaphor. Relativity establishes that there is no privileged frame: no single vantage point from which the universe is fully specified, since simultaneity and interval are frame-dependent and causal access is bounded by the light cone. Quantum mechanics goes further: prior to interaction, properties are not merely unknown but undefined, and the relational reading of quantum states makes this explicit — state descriptions are relative to the system doing the interacting [3]. Wheeler's formulation of the same intuition — that physical entities derive their existence from the resolution of binary alternatives by measurement — states the ontology in information-theoretic terms [4].

Information, then, is not something **about** reality. It is the name an observer gives to physics not yet resolved from where that observer stands. There is no second realm; there is physics, and there are the names attached to the parts of it a perspective has not yet reached.

3. Why Asymmetry and Entropy Are Identical, Not Analogous

The proportionality between information asymmetry and physical entropy is frequently treated as an analogy — a bridge between Shannon's domain [5] and Boltzmann's [6]. The ontology above implies it is not a bridge but an identity.

Entropy is observer-relative. It counts the microstates indistinguishable given the macrostate accessible to the observer; change the observer's resolution and the entropy changes. Jaynes's derivation of statistical mechanics from inference makes this explicit: thermodynamic entropy is the entropy of the probability distribution that expresses an observer's state of knowledge, subject to the constraints that observer can measure [7]. It was never a property of the system alone.

Information asymmetry has the same structure: it is the physics that has not collapsed within the observer's causal reach — what is hidden, undefined, or unresolved from a given vantage point. Also not a property of the system alone.

Two quantities, both measuring physics an observer has not resolved. They are proportional because they are the same measurement under different vocabularies. Landauer's principle supplies the physical hinge: resolving or erasing a bit dissipates a minimum energy of $k_B T \ln 2$, a bound belonging to thermodynamics rather than to engineering [8], confirmed experimentally [9]. If knowing costs energy, the informational and the thermodynamic are not two ledgers.

4. The One Who Points Is Always a Nervous System

The objection that should stop this argument is reference. If everything is physics, what performs the pointing? The word "market" is about markets and not about stones. A theory is about the world. That relation of *aboutness* appears to be precisely the abstract, non-physical entity the argument denies.

The answer proposed here: the one who points is never an abstract something. It is always a nervous system — evolved from the same matter — and it pays a thermodynamic cost to point. There is no free-floating reference anywhere in the universe. Every act of meaning is executed by a physical, evolved organism spending real energy to execute it. By Landauer's bound, that expenditure has a floor the Second Law does not waive.

Reference is therefore not an exception to physicalism; it is a physical event that dissipates energy — and the cost is the argument. If pointing were free, it would be abstract. Because it costs an organism something real, it is material. Meaning does not escape thermodynamics; meaning is thermodynamics, performed by biology.

This makes the nervous system the keystone where four otherwise separate concerns meet: **perspective** (the vantage point that creates asymmetry), **cost** (the metabolic and physiological wear of resolving), **inscription** (the writing of state into the organism), and **meaning** (the locus where physics comes to be about other physics). The claim is that perception, wear, inheritance, and meaning are not four subjects but one — an organism managing asymmetry at a cost — observed from four sides.

5. Geometry, Not Destiny — and What the Adjudications Say

This ontology yields a methodological prediction with empirical consequences.

The uncollapsed has geometry. It does not have destiny. The field of what has not yet occurred has structure — shape, constraint, curvature — but no settled outcome, because a settled outcome would already be collapsed. An instrument built to read the uncollapsed should therefore read its geometry and should be structurally unable to read its outcome.

The author's programme has tested instruments of exactly this kind — estimators of state on the Fisher–Rao statistical manifold, whose metric is fixed by invariance rather than chosen [10, 11] — and the adjudications are informative in both directions.

On the structural side, an irreducible fluctuation floor appears in human gaze and replicates across two independent laboratories at matched resolution to within 0.6%, while the same instrument, unmodified, finds no such floor in settled price series [1, 2]. Floors appear where a system estimates a future and vanish where one merely settles a past.

On the directional side, the results went against the framework's more ambitious readings, and were published as such. In a preregistered comparison, the geodesic anchor proved indistinguishable from a moving average in direction, and an informed mean-reverting baseline matched or beat the information-geometric barriers at equal width [12]. A separate preregistered test of a bridge from the gaze measure to autonomic state returned NOT ASSOCIATED, and the author's registered withdrawal was executed [13].

The directional null is the point of interest here. Under the ontology, direction *is* the outcome, and the outcome is potential rather than fact — there is no "which way" available to be read, only a field of possibles. A reader of the uncollapsed should read shape and go silent on destiny. The directional unpredictability repeatedly encountered is therefore not evidence against the framework: it is what the framework predicts. Stated honestly, the claim is narrower and stronger — these instruments read geometry, not destiny, and the containment they exhibit is distinguishability from uninformed nulls, not superiority over the cheapest informed explanation.

6. Deception Is Not a Bug. It Is the Prediction.

The deepest difficulty for any informational ontology is false reference. If meaning is the reduction of asymmetry about X by a nervous system, how can a symbol be about X and be wrong? A camouflaged moth is about bark and is not bark; a lure is about prey and is a trap.

Under this framework the difficulty inverts. Misrepresentation is not a malfunction of information but one of its oldest and most successful uses. Sensory systems are not passive windows; they are bidirectional. They have an input function — reducing the organism's own asymmetry by sampling the environment — and a signature function — managing other organisms' asymmetry through what the body broadcasts. Evolution tunes the second relentlessly: crypsis and disruptive coloration increase a predator's asymmetry; Batesian mimicry manufactures a false belief about threat [14]; aposematism is the honest counter-move, reducing a predator's asymmetry truthfully to mutual benefit.

A symbol can be about X and be false because being usefully false — making another organism's map diverge from the territory in one's favour — is itself a function selection acts upon. The map was never required to be true. Von Uexküll's *Umwelt* makes the same point from the perceptual side: an organism's world is not a one-to-one copy of its environment but the functionally relevant slice, optimized for action rather than accuracy [15].

A final turn returns this to thermodynamics. Zahavi's handicap principle holds that a signal is reliable when it is costly — when only a high-quality organism can afford to produce it [16]. Reliability, in biology, is purchased with energy. The boundary between honest and deceptive signaling is therefore a thermodynamic boundary, which is the identity of Section 3 written across the living world: the cost of a signal is the measure of its truth.

7. The Same Loss at Different Speeds

Viewed from a distance, the picture is one process at different rates. A participant who expends capital trading noise, and an organism that wears down under sustained load, lose for a structurally similar reason: both spend irreplaceable resources maintaining symbols over uncollapsed physics. The market absorbs the first and returns to its line. The organism absorbs its own and degrades in doing so.

This last statement is an interpretive extension, not a measured result, and is flagged as such: the empirical content of this programme is the floors, the estimating-versus-settling asymmetry, and the resolution-dependence of the residual, together with the adjudications listed in Section 5. Whether the

physiological analogue holds is the subject of preregistered work, and the framework's own nulls in that domain have been published at the volume of its confirmations [13].

What is claimed is the ontology and its two consequences: that reality and abstraction are physics in two states relative to an observer, that information is the name for the unresolved state, and that whoever measures is always a physical system paying a real cost — which is the argument that none of it was ever abstract.

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